

Statistical Mechanics

统计力学

Course Information

Lectures

- Monday 09:50–11:25
- Wednesday 08:00–09:35
- 新水利馆 331
- Exercise classes: time and place TBD

Office hours

- Monday 08:00–09:50
- 双清综合楼 A座 B522
- Email for appointments at other times.

Instructor and TAs

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- Teaching assistant: 王皓东
- hd-wang23@mails.tsinghua.edu.cn

Assessment

Component	Weight
Attendance (college requirement)	10%
Homework	20%
Midterm exam	35%
Final exam	35%

Attendance + Homework + Midterm + Final = 100%

- **Both the midterm and final exams are closed-book.**
- If AI helps with homework, understand the reasoning, then redo the problem on your own without AI or the solution.

References

Main reference

- David Tong, Statistical Physics
[lecture notes · free online](#)
- Lecture notes will be posted after each week's classes.

Further reading

- Mehran Kardar, Statistical Physics of Particles
- L. D. Landau and E. M. Lifshitz, Statistical Physics, Part 1

What is Statistical Mechanics?

- Statistical mechanics connects microscopic laws to the collective properties of matter.
- Probabilities of microstates determine macroscopic averages.

$$\langle A \rangle = \sum_s p_s A_s$$

s : a microstate p_s : its probability A_s : observable

- Why do gases exert pressure? Why do magnets lose order?

Topics of This Course

(already fixed by Qiuzhen College)

- Fundamentals of Statistical Mechanics

Microstates, entropy and ensembles

- Classical Gases

Partition functions and macroscopic observables

- Quantum Gases

Bose and Fermi statistics

- Classical Thermodynamics

Thermodynamic laws and potentials

- Phase Transitions

Order parameters, criticality and universality

Ensembles and Gases

Topic	Core ideas	Questions
Statistical foundations	Entropy Ensembles Partition functions	Which states are likely? How do we compute averages?
Classical gases	Ideal gas Equipartition Interactions	Where does pressure come from? When does the ideal gas fail?
Quantum gases	Bose-Einstein statistics Fermi-Dirac statistics Degeneracy	What changes for indistinguishable particles?

Thermodynamics and Phase Transitions

Topic	Core ideas	Questions
Thermodynamics	Laws of thermodynamics Thermodynamic potentials Response functions	What constrains heat, work and equilibrium?
Phase transitions	Order parameters Landau theory Critical phenomena	How does collective order emerge and disappear?

A theory is the more impressive the greater the simplicity of its premises, the more different kinds of things it relates, and the more extended its area of applicability. Therefore the deep impression that classical thermodynamics made upon me. It is the only physical theory of universal content which I am convinced will never be overthrown, within the framework of applicability of its basic concepts.

—Albert Einstein

一个理论的基本前提越简单，所联系的事物种类越多，适用范围越广，就越令人赞叹。因此，经典热力学给我留下了极其深刻的印象。它是唯一一个具有普遍性内容的物理理论，并且我坚信——在其基本概念的适用范围内——它永远不会被推翻。

——爱因斯坦

Other Mechanics

New formulations, limits and unification

- Reformulation: Newtonian, Lagrangian and Hamiltonian mechanics describe the same classical dynamics.
- Relativity: Newtonian mechanics is the low-speed limit ($v \ll c$) of special relativity.
- Quantum mechanics: A typical classical-limit criterion is $S \gg \hbar$, where S is an action scale.
- Gravity: General relativity has a Newtonian limit. A complete theory of quantum gravity remains open.
- Unification: Maxwell's electromagnetism leads to QED and, with the weak interaction, electroweak theory.

Statistical Mechanics

Statistical mechanics: from microscopic to macroscopic

- Microscopic input: Start from classical or quantum laws for many interacting degrees of freedom.
- Macroscopic predictions: Use probability to describe thermodynamics, fluctuations and collective behavior.
- Coarse-graining: Discrete models can lead to continuum fields, linking algebra, probability and analysis.
- Universality: Near continuous transitions, many details become unimportant. Symmetry and dimension still matter.
- Emergence: Collective phenomena include phase transitions, spontaneous symmetry breaking and superfluidity.

$$10^{23} \neq 1$$

- Consider a magnet with $N \sim 10^{23}$ spins. Each spin points up or down, encoded as 0 or 1.
- For reference: 1 mol contains 6.02×10^{23} spins.
- Storing one spin configuration requires N bits.
- 10^{23} bit $\sim 10^{22}$ B $\sim 10^{19}$ kB $\sim 10^{16}$ MB $\sim 10^{13}$ GB $\sim 10^{10}$ TB
- At an assumed SSD price of RMB 1000 per TB, the storage cost is $\sim$ RMB 10^{13} .
- 10 万个“小目标”
- For a quantum system of N spin-1/2 particles:

$$\dim \mathcal{H} = 2^N \sim 2^{10^{23}}$$

Arrow of Time

上邪（佚名）

上邪！我欲与君相知，
长命无绝衰。
山无陵，江水为竭，
冬雷震震，夏雨雪，
天地合，乃敢与君绝！

热律（DeepSeek）

热律！熵增不可逆，
时空永无衰。
热自冰传火，气自聚成埃，
老返童，水归山，
机永动，乃敢言此律乖！

More Is Different

Broken symmetry and the nature of the hierarchical structure of science.

P. W. Anderson

X
solid state or
many-body physics
chemistry
molecular biology
cell biology
•
•
•
psychology
social sciences

Y
elementary particle
physics
many-body physics
chemistry
molecular biology
•
•
•
physiology
psychology

• Reductionism 还原论

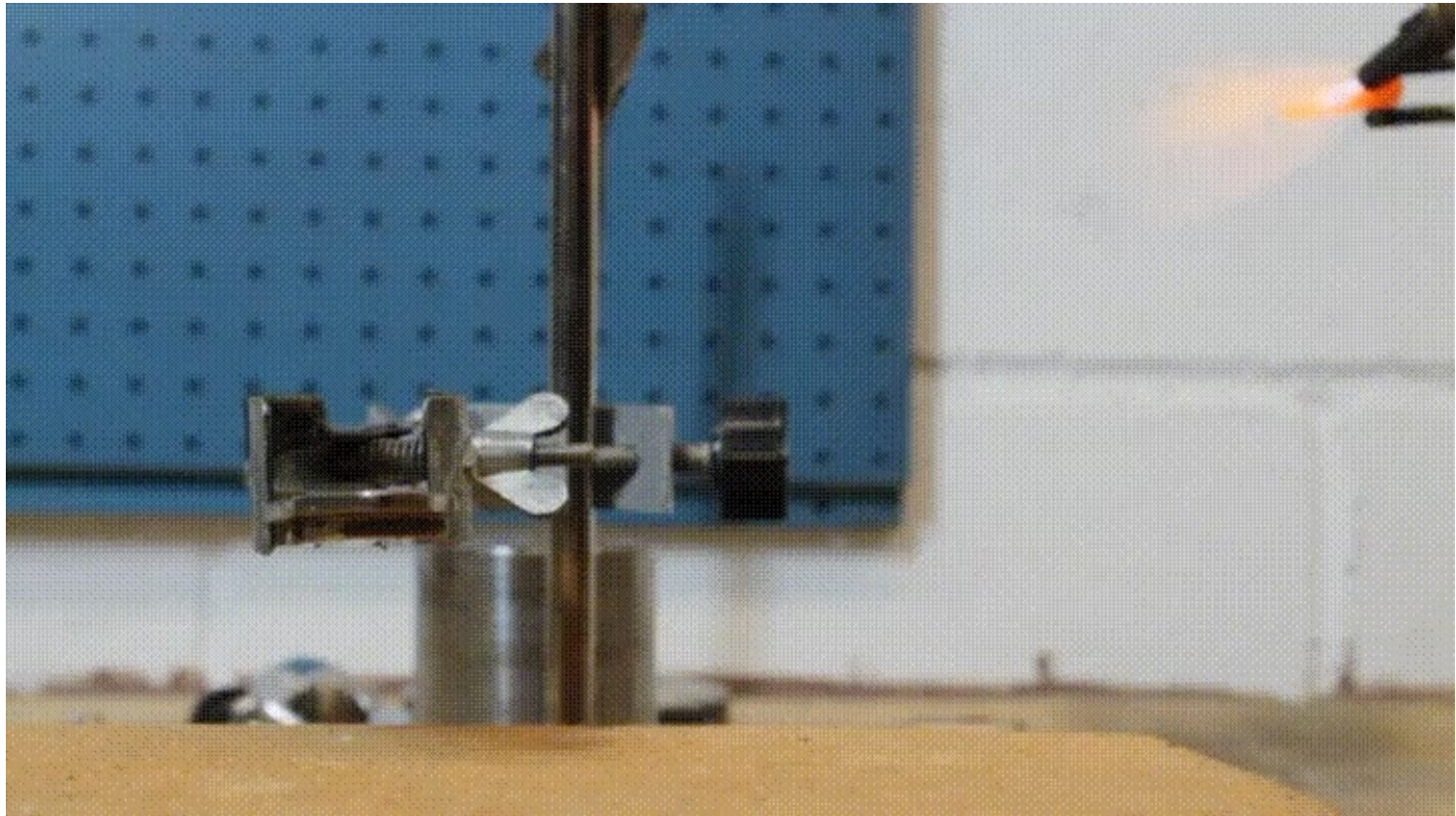


• Emergentism
演生 / 涌现 / 层展论



Magnetism at the Curie Temperature

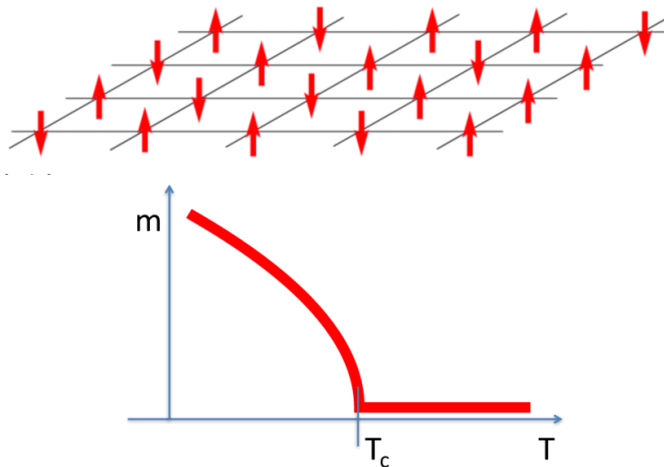
- Heating changes the magnetic response.



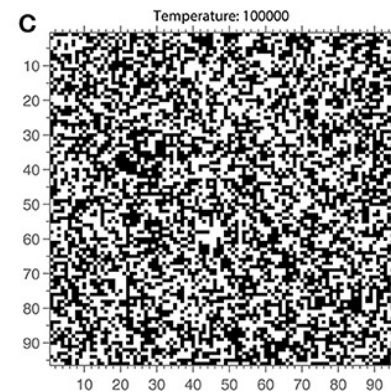
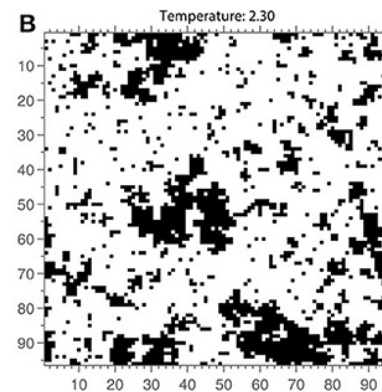
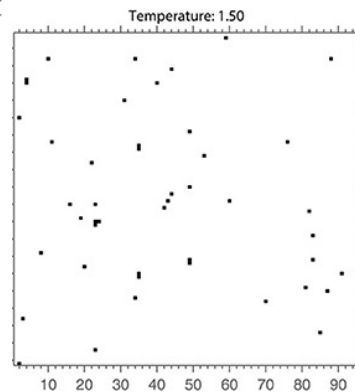
Ising model

$$\lim_{h \rightarrow 0} \lim_{N \rightarrow \infty} \langle B \rangle_{N,h} \neq \lim_{N \rightarrow \infty} \lim_{h \rightarrow 0} \langle B \rangle_{N,h} = 0$$

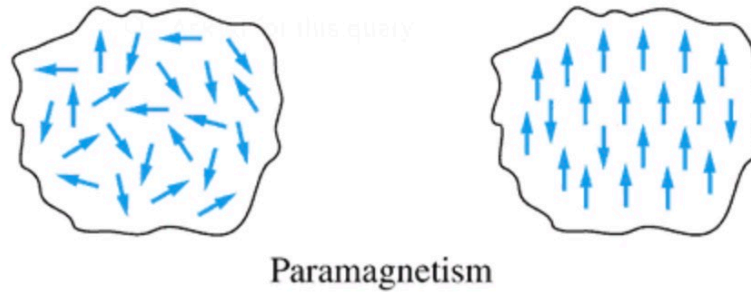
- Proposed by W. Lenz in 1920 $H = -J \sum_{\langle ij \rangle} s_i s_j$
- 1D: No phase transition by E. Ising in 1925
- 2D: Phase transition by L. Onsager in 1944
- 3D?



T_c : Curie temperature

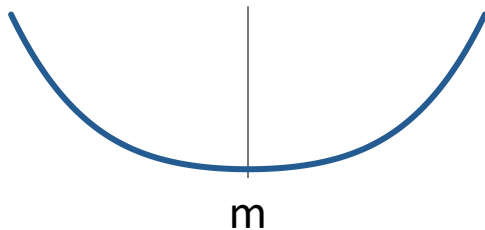


Spontaneous Symmetry Breaking

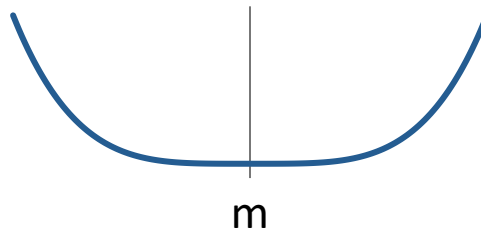


$$F(m) = a(T)m^2 + b m^4, \quad b > 0$$

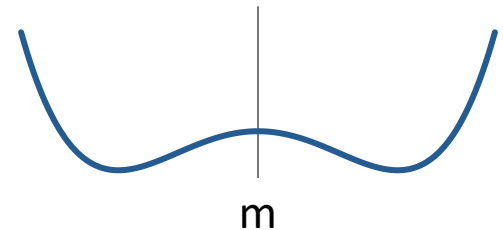
$T > T_c$



$T = T_c$



$T < T_c$



- At $h = 0$, changing m to $-m$ leaves F unchanged.
- At low T , a pure phase selects one minimum.

$$m = \lim_{h \rightarrow 0^+} \lim_{N \rightarrow \infty} \langle m \rangle$$

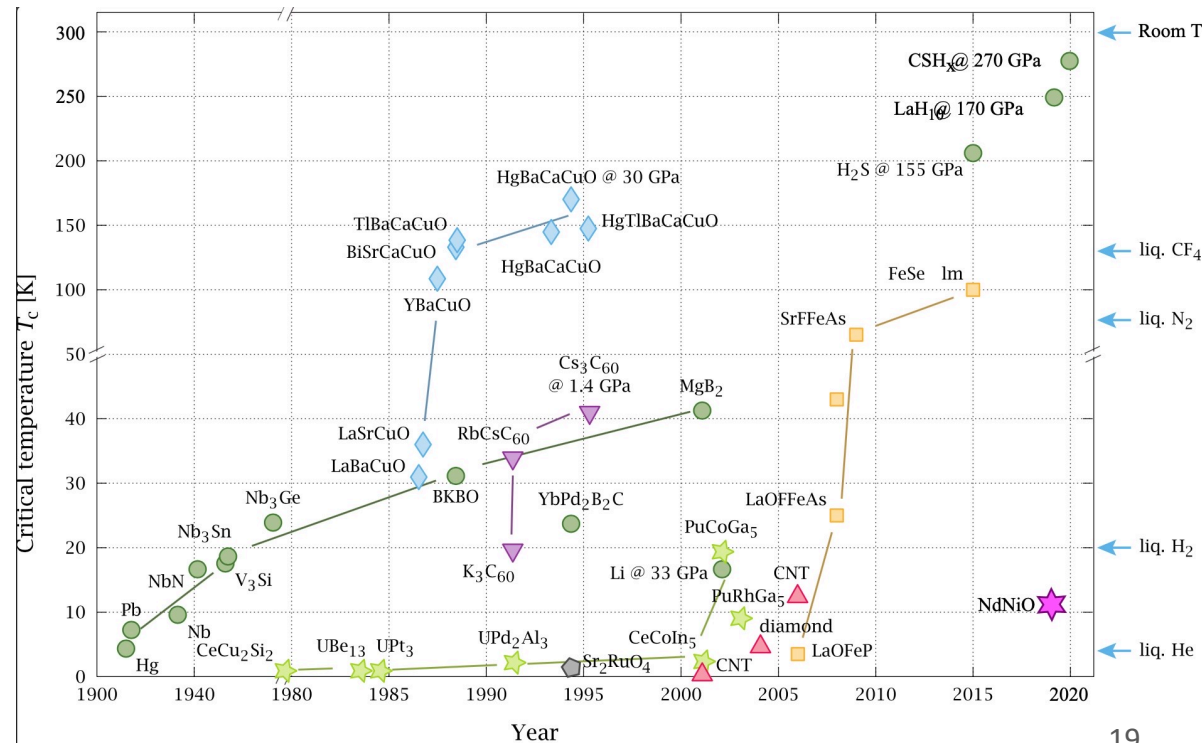
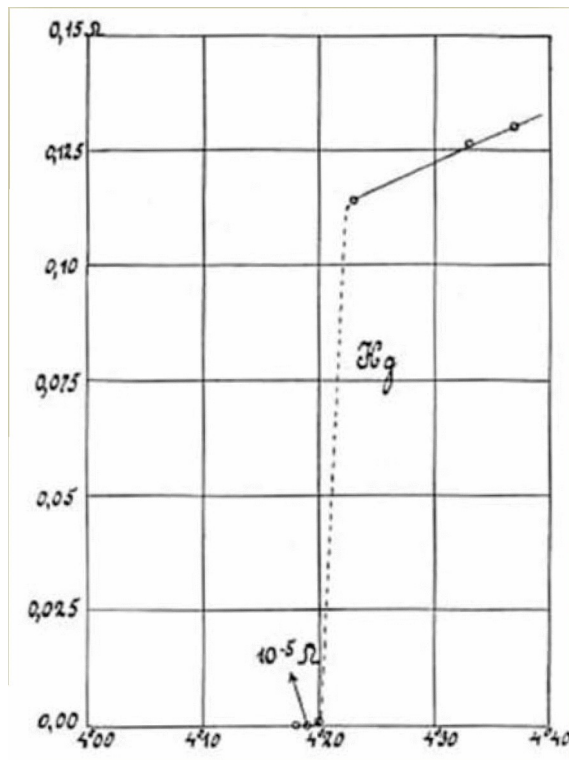
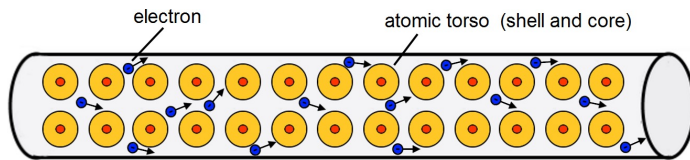
Superfluid (超流)

- Zero viscosity
- Spontaneous symmetry breaking of number conservation $U(1)$ symmetry



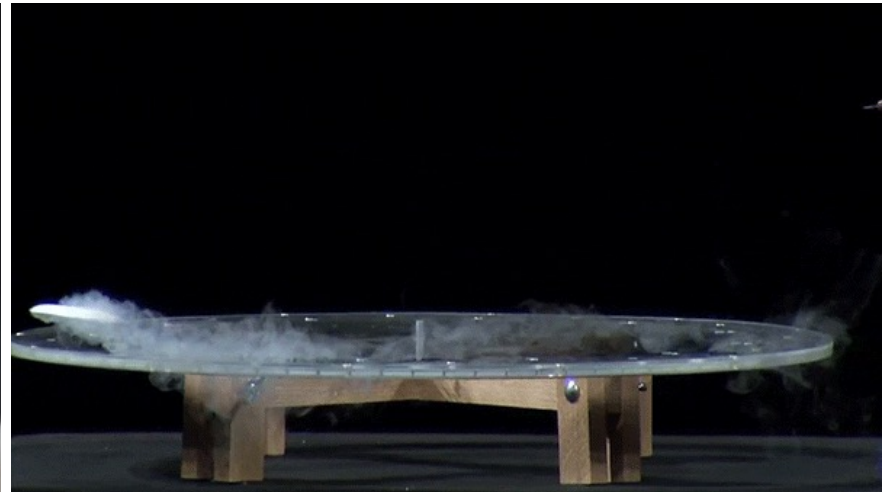
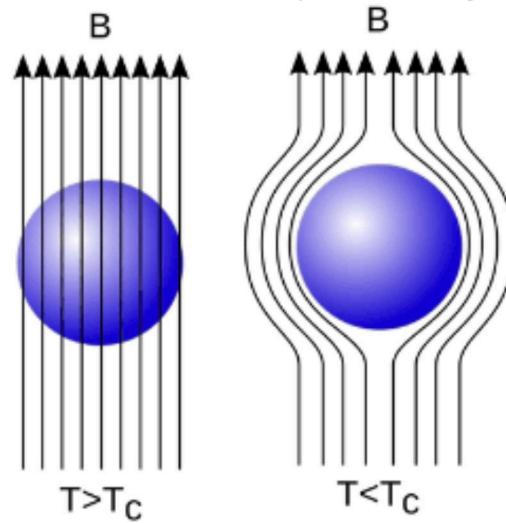
Superconductor (超导)

- Onnes 1911, mercury, $T < 4.15\text{K}$, zero resistance



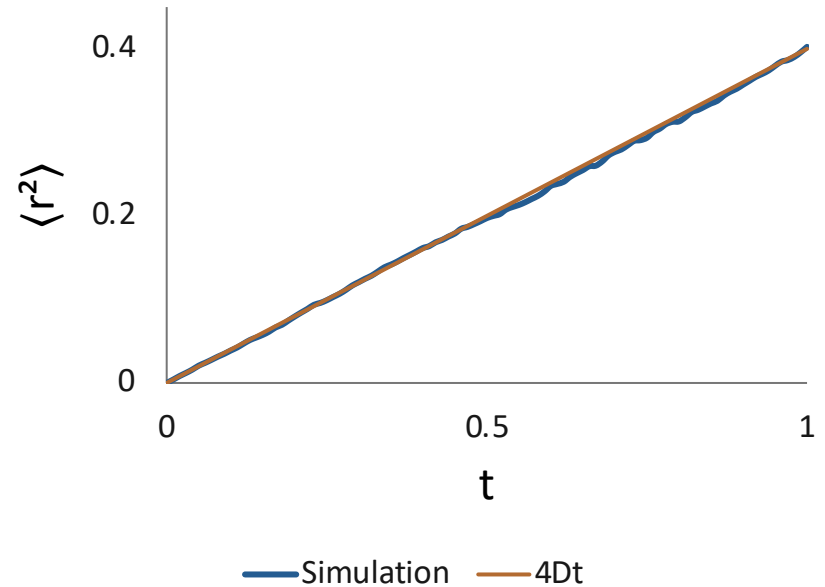
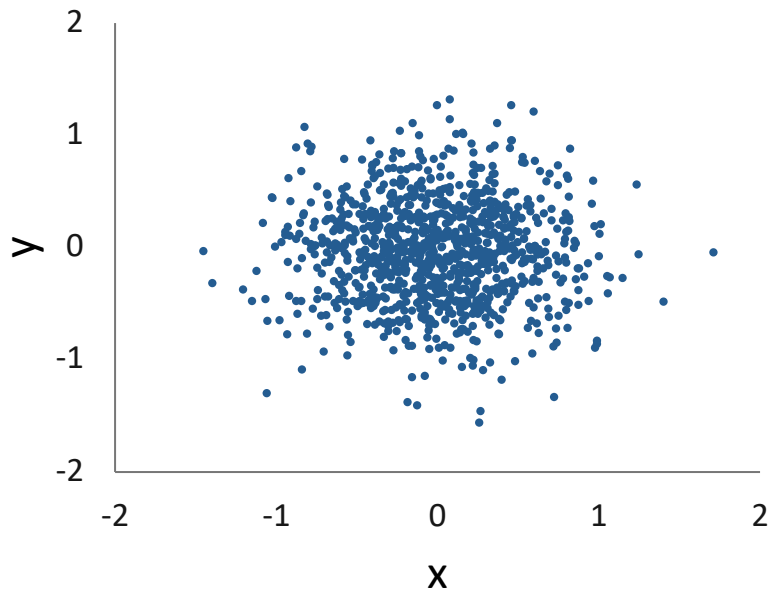
Meissner effect

- In the Meissner state, a superconductor expels magnetic flux.



Brownian Motion

- 2D random walks of 1000 particles, starting at the origin



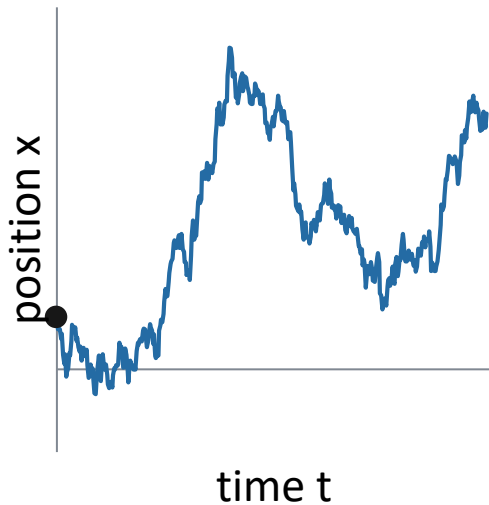
$$\langle r^2(t) \rangle = 4Dt$$

- Individual paths are irregular.
- Ensemble averages follow a law.

Brownian Motion and Dimension

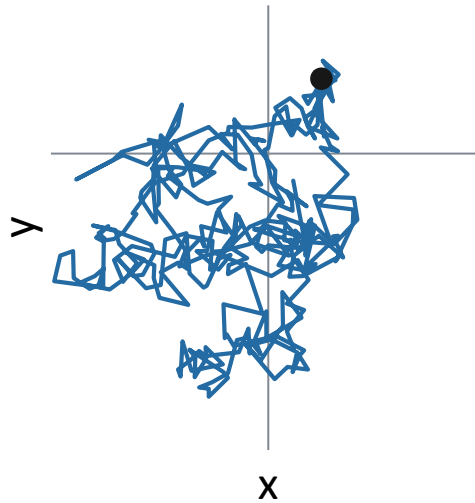
- Drunken birds may not return home.

1D



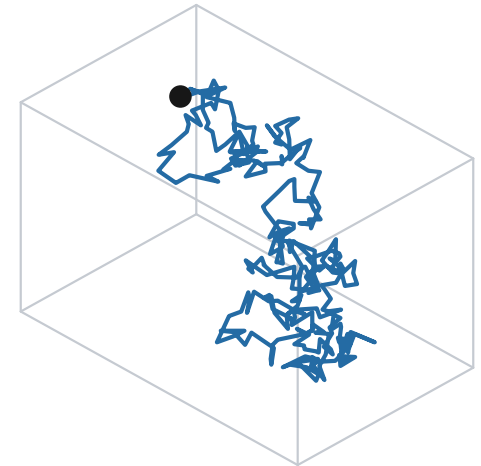
- Reaches home with probability 1

2D



- Reaches home with probability 1

3D



- May never reach home

Return Probability and Dimension

Simple symmetric walk on $\mathbb{Z}^d$: $p_{2n} = P(S_{2n} = 0)$

- **1D: count n left steps and n right steps**

$$p_{2n} = \frac{(2n)!}{2^{2n}(n!)^2} \sim \frac{1}{(\pi n)^{1/2}}$$

- **1D: diffusion length $\ell \sim \sqrt{n}$**

$$p_{2n} \sim \frac{1}{\ell} \sim n^{-1/2}$$

- **Higher dimensions: approximately independent directions**

$$2D: p_{2n} \sim (n^{-1/2})^2 = n^{-1} \qquad 3D: p_{2n} \sim (n^{-1/2})^3 = n^{-3/2}$$

Long-time scaling, ignoring constant prefactors.

- **1D / 2D: recurrent**
- **3D: transient**

$$\sum_{n=1}^{\infty} n^{-d/2} = \infty$$

$$\sum_{n=1}^{\infty} n^{-3/2} < \infty$$

$$G(0) = \frac{1}{(2\pi)^3} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \frac{dk_x dk_y dk_z}{1 - \frac{1}{3}(\cos k_x + \cos k_y + \cos k_z)}$$

$$P_{\text{return}} = 1 - \frac{1}{G(0)} \approx 0.3405$$

Statistical Mechanics and Mathematics

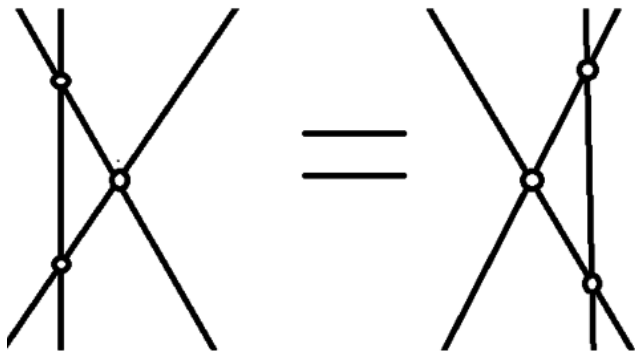
Selected connections, ordered by Fields Medal year

Year	Medalist	Physical setting	Mathematical contribution
1990	Vaughan Jones	Solvable lattice models	Operator algebras and knot invariants
2006	Wendelin Werner	2D random curves	SLE and Brownian geometry
2010	Stanislav Smirnov	2D percolation and Ising models	Conformal invariance of scaling limits
2010	Cédric Villani	Gases and plasmas	Boltzmann relaxation and nonlinear Landau damping
2014	Martin Hairer	KPZ and noisy fields	Regularity structures for singular stochastic PDEs
2022	Hugo Duminil-Copin	Ising and percolation	Theory of phase transitions
2026	Yu Deng	Gases and wave turbulence	Derivation of Boltzmann and wave kinetic equations

Integrable Models and Knot Invariants

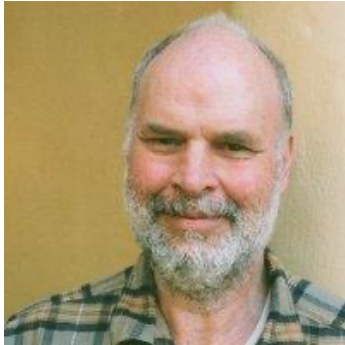
Yang–Baxter equation

$$(\check{R} \otimes \mathbf{1})(\mathbf{1} \otimes \check{R})(\check{R} \otimes \mathbf{1}) = (\mathbf{1} \otimes \check{R})(\check{R} \otimes \mathbf{1})(\mathbf{1} \otimes \check{R})$$



- Local consistency supports commuting transfer matrices.
- Some solutions connect to quantum groups, braid representations and topological invariants.

Subfactors and Knots



Vaughan Jones

Fields Medal 1990

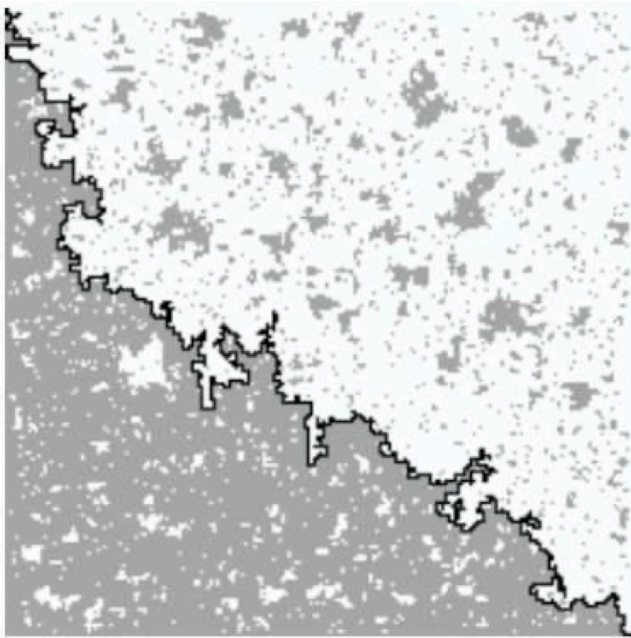
- Official laudatio (excerpt): “In 1984 Jones discovered an astonishing relationship between von Neumann algebras and geometric topology.”

- Subfactor index theorem: only discrete index values occur below 4.
- Jones polynomial: a new invariant of knots and links from braid representations.
- Temperley–Lieb algebras also describe solvable lattice models.
- Kauffman’s state sum: a partition function

$$\langle D \rangle = \sum_s A^{a(s)-b(s)} d^{n(s)-1}, \quad d = -A^2 - A^{-2}$$

Criticality, CFT and Probability

- 2D criticality: long-range fluctuations and scale invariance



Conformal field theory (CFT)

- Virasoro algebra and vertex operator algebras

Schramm–Loewner evolution (SLE)

- Critical Ising spin interfaces: $\kappa = 3$
- Critical Ising FK interfaces: $\kappa = 16/3$

- Under suitable conditions, 2D scale invariance implies conformal invariance.

Brownian Geometry and SLE



Wendelin Werner

Fields Medal 2006

- Official citation: “For his contributions to the development of stochastic Loewner evolution, the geometry of two-dimensional Brownian motion, and conformal field theory”
- With Lawler and Schramm: exact Brownian intersection exponents.
- Planar Brownian frontier: fractal dimension $4/3$.
- SLE links random critical interfaces with conformal field theory.

$$\partial_t g_t(z) = \frac{2}{g_t(z) - \sqrt{\kappa} B_t}$$

Conformal Invariance



Stanislav Smirnov

Fields Medal 2010

- Official citation: “For the proof of conformal invariance of percolation and the planar Ising model in statistical physics.”
- Critical site percolation on the triangular lattice: Cardy’s crossing formula.
- Planar Ising model: discrete holomorphic observables establish conformal invariance.
- Scaling limits: percolation interfaces follow SLE_6 , while Ising spin interfaces follow SLE_3 .

Entropy and Relaxation



Cédric Villani

Fields Medal 2010

- Official citation: “For his proofs of nonlinear Landau damping and convergence to equilibrium for the Boltzmann equation.”
- Boltzmann equation: quantitative rates of convergence to equilibrium.
- With Mouhot: nonlinear Landau damping by phase mixing in collisionless plasmas.
- Optimal transport: geometric and functional inequalities.

$$\partial_t f + v \cdot \nabla_x f = Q(f, f)$$

$$\mathcal{H}(f) = \int f \log f \, dx \, dv, \quad \frac{d\mathcal{H}}{dt} \leq 0$$

The KPZ Universality Class

$$\partial_t h = \nu \nabla^2 h + \frac{\lambda}{2} (\nabla h)^2 + \eta$$

- For interface height h : smoothing, local growth and random noise.



Kardar–Parisi–Zhang (1986)

- Stochastic PDEs connect growth phenomena with probability and analysis.

Singular Stochastic Equations



Martin Hairer

Fields Medal 2014

- Official citation (excerpt): “... for the creation of a theory of regularity structures for such equations.”
- Regularity structures (2014): a general framework for singular stochastic PDEs.
- KPZ (2013): a pathwise solution via renormalization.
- The theory also treats the dynamical Φ^4_3 model.

Phase Transitions



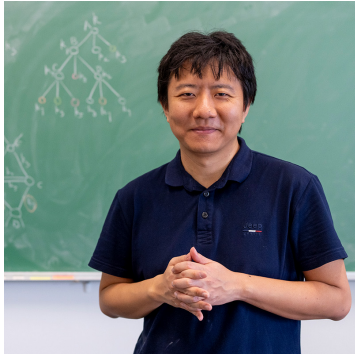
Hugo Duminil-Copin

Fields Medal 2022

- Official citation: “For solving longstanding problems in the probabilistic theory of phase transitions in statistical physics, especially in dimensions three and four.”
- Sharpness: exponential decay in the subcritical regime.
- 3D Ising model: spontaneous magnetization vanishes continuously at criticality.
- With Aizenman: Gaussian scaling limits for critical 4D Ising and lattice ϕ^4 models.

$$m(T_c) = 0$$

Microscopic Dynamics



邓煜

Fields Medal 2026

- Official citation (excerpt): “... the rigorous derivation of the Boltzmann equation from hard-sphere dynamics for rarefied gases, the derivation of wave kinetic equations from nonlinear dispersive systems, and probabilistic approaches to nonlinear Schrodinger dynamics.”
- With Hani and Ma: derive the Boltzmann equation from hard-sphere dynamics.
- With Hani: derive wave kinetic equations for weakly nonlinear waves.
- With Nahmod and Yue: random-data methods for nonlinear Schrödinger equations.
- Boltzmann derivation: within the regular solution’s lifespan.

Frontiers and Open Questions

Thermalization

- When do isolated quantum systems reach thermal equilibrium?

Nonequilibrium dynamics

- Which principles organize driven and dissipative matter?

Criticality in three dimensions

- How can we determine universal data beyond exact solutions?

Correlated quantum matter

- What mechanisms produce unconventional superconductivity?

Disorder and gravity

- What controls finite-dimensional spin glasses and black-hole entropy?

.....

Ludwig Boltzmann, who spent much of his life studying statistical mechanics, died in 1906, by his own hand. Paul Ehrenfest, carrying on the work, died similarly in 1933. Now it is our turn to study statistical mechanics.

— David Goodstein, *States of Matter*